FEAT-ZD-48600 · DE-03 Database changes · Completed database changes

Data Engineering agent · item task.db_changes · source hobasa-data-engineering-agent/outputs/FEAT-ZD-48600/task.db_changes.md · approved by Data engineer sim.data_engineer

task.db_changes: FEAT-ZD-48600

Completed database changes for FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is DE-03 Database changes (attempt 1).

Result: this feature needs no database changes, so there is nothing left to do. The step is complete and nothing has to be applied.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DE-03 Database changes (attempt 1)	record:brief
Workbook action	Check the required database changes are completed	record:brief
Owner / gate	Data engineer / no gate: automated checks, independent judge, sampled human evaluation	record:brief
Inputs	feat.tasks (b0fd4d2), feat.design (722a98f)	record:brief
Data Engineering tasks in scope	TSK-01 (STORY-01, `paid_date` in `to_model`), TSK-02 (STORY-02, `delays_by_vendor`)	feat.tasks (section 1)
Repository and runtime	ap_pipeline, Python 3.9 standard library only	feat.tasks (section 1); feat.design (DD-1, B1)

1. What the design says about storage

Check	Finding	Source
Storage in scope	The design's data boundary covers one new record key `paid_date`, one new reject reason `'paid_date'`, an in-memory delay history, risk map and overdue rows, and CT-1 JSON. It lists "Storage, database, schema migration" as **out of this design**	feat.design (A1, Boundary "Data")
New technology or storage	DD-1: no new repository, service, runtime, library or storage	feat.design (A7, DD-1)
Purity of new functions	NF-10: every new function is pure and in-memory, with no I/O, no network and no mutation of inputs	feat.design (A5, NF-10)
Platform trigger	Storage or a network call from the libraries would reopen DES-01. The design triggers none of these	feat.design (A8)
Invoice record shape	DC-1 is a plain Python `dict`, not a class; `paid_date` is added as a key (a `date` or `None`)	feat.design (A3 DC-1; B1.3)
Delay history	DC-2 is an in-memory `Dict[str, List[int]]`	feat.design (A3 DC-2; B1.3)

2. What the Data Engineering tasks change

Task	Files in budget	Database object touched	Source
TSK-01	`src/ap_pipeline/transform.py`; `src/ap_pipeline/__init__.py` (version step only); `tests/test_pipeline.py`	None	feat.tasks (TSK-01)
TSK-02	New `src/ap_pipeline/delays.py`; new `tests/test_delays.py`	None	feat.tasks (TSK-02)

TSK-01's "must not change" list keeps `ingest.py`, `dq.py`, `pipeline.py` and `schedule.json` as they are (feat.tasks TSK-01; feat.design B1.1). No task in feat.tasks lists a DDL file, a migration script or a table.

3. What the repository holds (code read)

Item	Finding	Database access	Source
Repository	ap_pipeline has no database, has no migrations and needs none	None	tool:repo-access
`ingest.parse_feed`	Parses CSV text in memory with `csv.DictReader` over `io.StringIO`	None	file:ap_pipeline/src/ap_pipeline/ingest.py
`transform.to_model`	Builds plain `dict` records inside a `try` block and rejects rows on `(KeyError, ValueError, InvalidOperation, AttributeError)`. TSK-01 adds the `paid_date` parse after that block	None	file:ap_pipeline/src/ap_pipeline/transform.py; feat.tasks (TSK-01)
`dq.quality_report`	Counts rows, loaded and rejected, and lists sorted `reject_reasons`, so the new reason `'paid_date'` appears without any change to this function	None	file:ap_pipeline/src/ap_pipeline/dq.py; feat.design (HC-3)
`pipeline.run`	`to_model(parse_feed(csv_text))`, then `quality_report`. Returns values and writes nothing	None	file:ap_pipeline/src/ap_pipeline/pipeline.py
`schedule.json`	daily, 02:00, Asia/Kolkata; no change	None	file:ap_pipeline/src/ap_pipeline/schedule.json; feat.design (B1.1)
`__init__.py`	`__version__ = "0.4.0"`; version step only (TSK-01)	None	file:ap_pipeline/src/ap_pipeline/__init__.py; feat.tasks (section 2)

A search of the ap_pipeline checkout for database terms (`sqlite`, `sql`, `database`, `migrat`, `cursor`, `connect()`) found no matches. Feed v2, which has no `paid_date` column, must still load. TSK-01 handles the optional column inside `to_model`, so no schema has to change (feat.tasks TSK-01; feat.design NF-4, B1.4).

4. Database change register

ID	Change	Required	Status	Source
—	No database, table, column, index, view or migration change	No	Not applicable: nothing to apply or roll back	feat.design (A1, DD-1, NF-10); feat.tasks (TSK-01, TSK-02); tool:repo-access

None of these apply: a rollback plan, migration scripts, an environment promotion order or post-change validation queries.

5. Effect on later Data Engineering steps

- DE-05 (Data Model Changes): the logical Invoice record gains `paid_date` (a `date` or `None`), and there is a new reject reason `'paid_date'`. These are changes to the registered data model, not to a physical database. They belong to DE-05, not to this step (feat.design B1.3, DC-1).
- DE-08 (Coding): this step adds no database prerequisite to TSK-01 or TSK-02.

6. Decisions and memory checked

Decisions D-2026-031, D-2026-044 and D-2026-050 and lessons L-007 and L-011 cover day counts, Decimal amounts and leaving paid invoices out. None of them needs storage, so none conflicts with a no-database outcome (record:memory).

7. Open questions (non-blocking)

ID	Question	Interim answer	Owner (proposed)
DBQ-1	Carried from feat.design DQ-1: no input names the production component that calls the stages and carries CT-1. A database change would only be needed if that host later stored records, overdue rows or risk levels	None in this feature. Adding storage would reopen DES-01 (feat.design A8)	Architect

8. Validation check

Check	Result	Source
Required database changes identified	None required	feat.design (A1, DD-1); tool:repo-access
Required database changes completed	Complete (nothing to apply)	tool:repo-access
No production or source-system data used	Pass: only the step's inputs, the repository code and the tool view were used; no source system was queried	record:brief
Conflict with an approved output	None	feat.design; feat.tasks
Blocking questions	None	—

[Tool result] [ap_pipeline] No database in this repository's architecture; no migrations required